# PLAN-10 - Risks, Dependencies, and Architecture Decisions

POS-DGFY - Global Location Switching | Plan v1.0 | Revision 2026-09-27

| Field | Value |
|---|---|
| Project | POS-DGFY - Global Location Switching |
| Plan version | 1.0 |
| Section | PLAN-10 - Risks, Dependencies, and Architecture Decisions |
| Revision date | 2026-09-27 |
| Plan status | NOT EXECUTION READY |
| Purpose | Make the execution dependency graph, effort, risks, and structural decisions explicit. |
| Dependencies | PLAN-01 through PLAN-09; repository phase ledger; architecture governance. |

## Dependency graph

```
Phase 354 contract and discovery
-> scope matrix and permission matrix
-> Phase 355 backend enforcement
-> Phase 356 shared context and UI
-> module integration and report/audit validation
-> Phase 357 hardening
-> pilot rollout
```

### Parallelizable work

- The module-scope inventory can run in parallel with UX wireframes after the contract boundaries are fixed.
- API authorization tests and UI selector tests can be prepared in parallel once requirement IDs are approved.
- Migration rehearsal can begin after schema ownership is confirmed, without enabling production writes.

### Risk register

| ID | Risk | Likelihood / impact | Mitigation and contingency |
|---|---|---|---|
| RISK-001 | Client or stale request mixes locations. | Medium / High | Context versioning, aborts, server scope checks; disable feature if leakage appears. |
| RISK-002 | Pending offline work is lost or assigned incorrectly. | Medium / High | Fail closed before switch; test reconnect and recovery. |
| RISK-003 | Voucher/price/affiliate scope is guessed. | High / High | Phase 354 schema and route audit; keep plan NOT EXECUTION READY. |
| RISK-004 | Strict terminal binding conflicts with branch switching. | Medium / High | Require destination enrolled terminal; document operator flow. |
| RISK-005 | Historical reports change after configuration switch. | Low / High | Immutable transaction snapshots and report regression tests. |
| RISK-006 | Dirty working tree contaminates implementation. | High / Medium | Use a clean dedicated worktree and baseline inventory. |
| RISK-007 | Phase ledger numbering conflicts with location docs. | Medium / Medium | Reconcile ledger before Phase 355. |
| RISK-008 | Global selector becomes an unsafe second switch path. | Medium / High | Selector dispatches only to governed Shift workflow. |

## Architecture decisions

| ADR | Decision | Sections |
|---|---|---|
| ADR-001 | Global entry point, Shift-owned operating transition, separate viewing context. | PLAN-03, PLAN-06 |
| ADR-002 | Shared catalog identity with location-specific operational state. | PLAN-03, PLAN-04, PLAN-07 |
| ADR-003 | Backend-derived scope and atomic/idempotent transition. | PLAN-05, PLAN-08, PLAN-09 |
| ADR-004 | Pilot and feature-flag rollout with additive migration only. | PLAN-07, PLAN-09 |

## Relative effort and feasibility

| Area | Effort | Reason |
|---|---|---|
| Discovery and contract | MEDIUM | Existing foundations reduce greenfield work; module scope is incomplete. |
| Backend enforcement | HIGH | Financial, inventory, authorization, and concurrency boundaries. |
| Frontend context and UX | MEDIUM | Existing state exists but needs central coordination and responsive UI. |
| Migration | UNKNOWN to HIGH | Depends on voucher, price-list, affiliate, and settings schema audit. |
| Testing and rollout | HIGH | Requires cross-module, security, migration, simulation, and recovery evidence. |